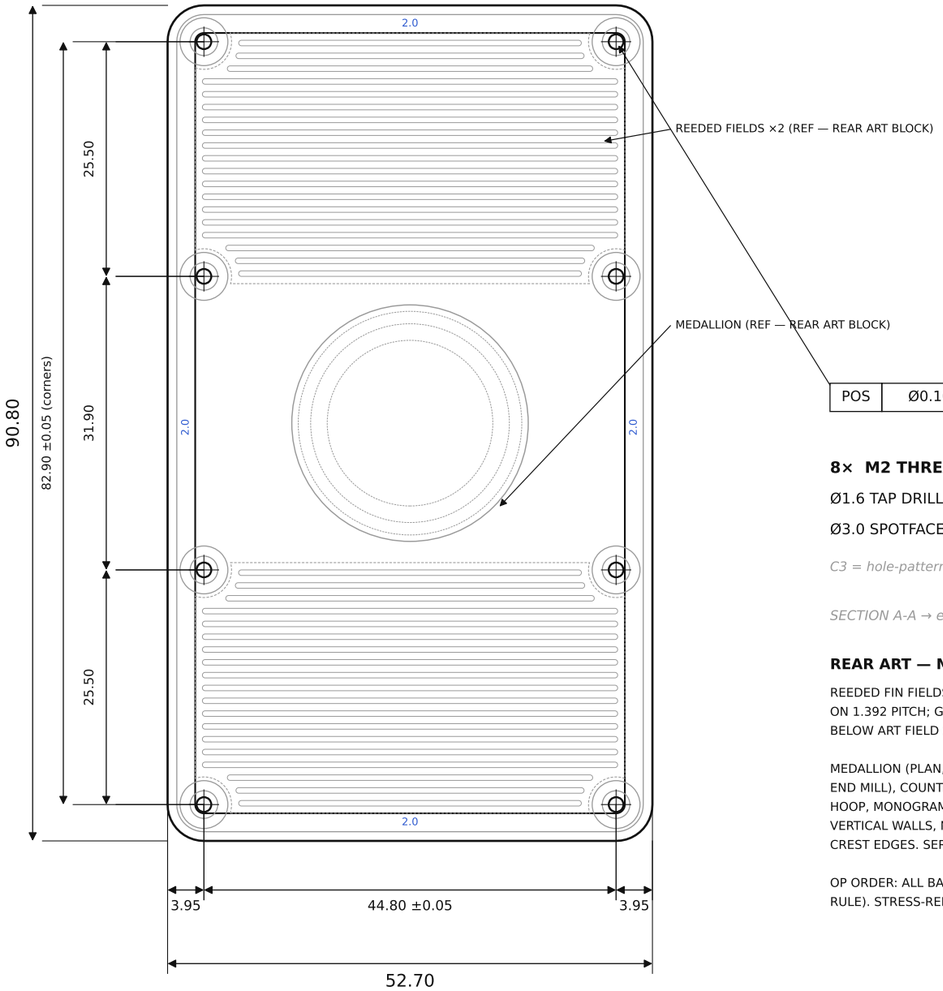
BACK FACE (outer) SCALE 1.5:1



POS	Ø0.10 (M)	A	B

8× M2 THREAD, THRU

Ø1.6 TAP DRILL • TAP FROM BACK FACE

Ø3.0 SPOTFACE, BACK (depth per Detail B)

C3 = hole-pattern pitch ±0.05   C4 = M2 thread

SECTION A-A → edge profile   DETAIL B → corner boss

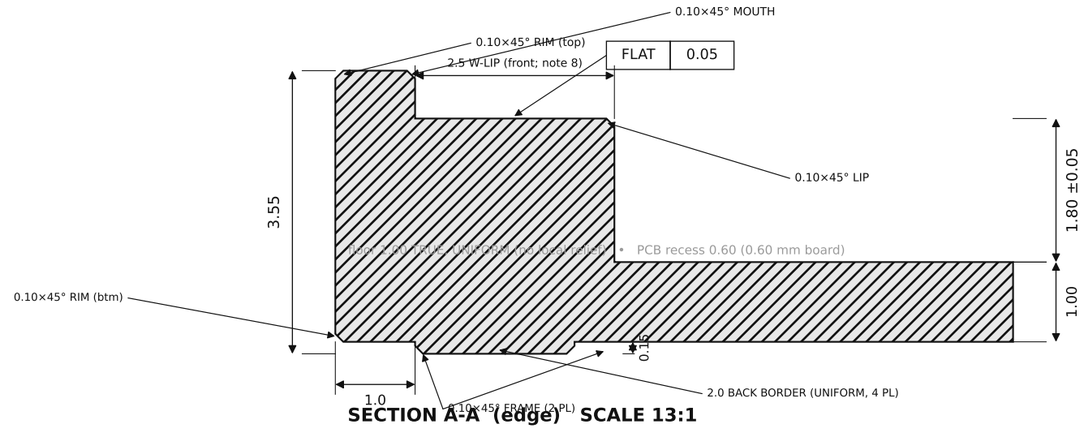
REAR ART — MACHINED-IN GEOMETRY (REFERENCE; STEP GOVERNS)

REEDED FIN FIELDS ×2 (PLAN, dashed zones): 19 RIBS/FIELD, 0.60 WIDE ON 1.392 PITCH; GROOVES Ø0.6 END-MILL ONLY (0.79 WIDE), VALLEY 0.60 BELOW ART FIELD (0.40 WEB); RIB TOPS +0.10 — 0.05 UNDER BEARING PLANE.

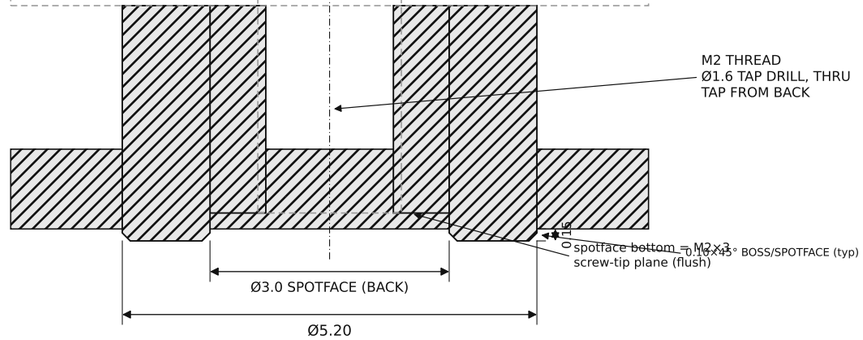
MEDALLION (PLAN, circles): Ø25.7 COIN SUNK 0.45 BELOW ART FIELD (Ø0.4 END MILL), COUNTERS/REST 0.25 (Ø0.3 REST PASS); CRESTS (RING TEXT, RIM, HOOP, MONOGRAM, SERIAL) LEFT AT THE FRAME'S BEARING PLANE (−0.15), VERTICAL WALLS, NO DETACHED ISLAND UNDER 0.55. DO NOT EDGE-BREAK CREST EDGES. SERIAL (“No 001”) IS PER-UNIT VARIABLE TEXT.

OP ORDER: ALL BACK-FACE FEATURES BEFORE CAVITY HOLLOW (SOLID-STOCK RULE). STRESS-RELIEVED STOCK — STATE CONDITION ON QUOTE.

SECTION A-A (edge) SCALE 13:1



C1 = cavity depth   C2 = PCB-rest-plane flatness   (back face down; PCB drops in from top)
PCB + M2×3 BRASS SCREW — REF, NOT THIS PART



DETAIL B (corner boss) 13:1

CRITICAL DIMENSIONS — C1 & C3 TOLERANCED ±0.05 (MARKED ON VIEWS); ALL OTHER FEATURES PER ISO 2768-1 (MEDIUM)

C1	Cavity depth (boss-top plane → cavity floor)	1.80 ±0.05
C2	PCB-rest plane flatness (lip + 8 bosses)	FLAT 0.05
C3	8× mounting holes (x 44.80; y rows 3.0/28.5/60.4/85.9)	±0.05 (linear)
C4	Mounting holes — tapped, thru, from back	M2 / Ø1.6 drill

GENERAL TOLERANCES (UNLESS OTHERWISE NOTED, mm)

LINEAR & HOLE-POSITION DIMS	**ISO 2768-m**
CAVITY DEPTH C1 (SECTION A-A)	**1.80 ±0.05**
HOLE-PATTERN PITCH C3 (8 HOLES)	**x 44.80 / y 25.5-31.9-25.5 ±0.05**
PCB-REST FLATNESS C2	**FLAT 0.05**

NOTES

1. MATERIAL: TITANIUM Gr5 (TC4) = Ti-6Al-4V GRADE 5.
2. FINISH: BEAD-BLAST MATTE, ALL OVER. NO MARKING SERVICE — THE REAR ART (REEDED FIELDS + MEDALLION) IS MACHINED-IN, IN THE STEP (SEE REAR ART BLOCK). BEARING PLANE (FRAME + MEDALLION CRESTS) HAND-LAPPED BY CUSTOMER AFTER BLAST — NO SHOP ACTION.
3. GENERAL TOLERANCE PER ISO 2768-1 (MEDIUM). C1-C4 TOLERANCED AS LISTED; C1 & C3 = ±0.05. DATUMS: A = LEFT EDGE, B = BOTTOM EDGE, C = PCB-REST PLANE.
4. BRACE REGISTRATION: THE RESIN H-BRACE REGISTERS TO THIS SHELL BY FITMENT - ITS 4 OUTBOARD RAILS + THE COMPONENT POCKETS + THE BOARD PRESS-FIT. NO LOCATOR PILLARS; THE CAVITY FLOOR IS A FULL 1.00 EVERYWHERE.
5. EDGE BREAKS - ALL 0.10x45°, FELT NOT SEEN, MODELED. CALLED OUT ON SEC A-A / DETAIL B: RIM = outer rim (top & bottom); MOUTH = recess mouth (around PCB); LIP = inner (cavity-side) lip edge; FRAME = proud back-frame bottom edges; BOSS = boss + spotface bottom edges (Detail B). BREAK ALL OTHER EXPOSED EDGES 0.10x45°. CONCAVE JUNCTION CORNERS (BOSS
6. FLOOR IS A TRUE 1.00, UNIFORM ACROSS THE CAVITY, AT THE ~1.0 Ti MIN-WALL GUIDANCE SO IT ALSO CLEARS ALUMINIUM / COPPER / STAINLESS.
   CAVITY 1.80 +-0.05 -> 1.75 WORST-CASE, MINUS WS17 1.70 MAX (DATASHEET CASE WS17 HEIGHT) = 0.05 NON-CONTACT; THE BRACE + SOLAR-CELL SANDWICHES CARRY THE BOARD.
7. NO U7 RELIEF POCKET. U7 IS THE 0.90 mm MB85RC512TY DFN-8, WELL UNDER THE 1.70 CAP-LIMITED CAVITY, SO THE FLOOR IS UNIFORM 1.00. (A POCKET EXISTED FOR A 1.75 SOIC-8 THE v4 BOARD DOES NOT CARRY.)
8. FRONT SUPPORT LIP IS ASYMMETRIC (SECTION A-A): WEST 2.5 / NORTH 2.0 / SOUTH 2.0 FOR PCB RIGIDITY; EAST 1.0 CLEARING THE JP1/TP1 PADS, THE NFC COIL (A GROUNDED LIP WOULD DETUNE IT), AND C7 (x49.55, THE ONE EAST-EDGE PART LEFT AFTER v4 REMOVED THE Q1/U4/R7/R9 CLAMP CLUSTER + D9/D10/D11); WIDENING TO 2.5 ONLY AT THE SOUTH END (y0-10, CLEAR C
9. NO INTERNAL RIBS OR SUPPORT POSTS: THE RESIN BRACE (SEPARATE PART) CARRIES CENTER SUPPORT. A PCB LAYOUT CHANGE = A BRACE REPRINT, NOT A SHELL RE-MACHINE.
10. PCB RECESS = 0.60 DEEP (RECEIVES THE 0.60 mm BOARD; SLIP FIT, NOT A PRESS FIT). 3D STEP GOVERNS ALL GEOMETRY; STL NOT FOR CNC.
11. PART = BACK-SHELL ONLY. PCB, RESIN BRACE, AND 8× M2×3 BRASS SLOTTED-CHEESE SCREWS (HEAD Ø3.8) SUPPLIED SEPARATELY.

SOLAR-GLOW DRH v3.0 — Ti BACK-SHELL (0.6mm-BOARD DUMB BOX)

DWG solar-glow-drh-v3_0-backshell-0p6b-brace	REV v3.0
MATERIAL Ti Gr5 (TC4)	FINISH BEAD-BLAST + LAP (CUSTOMER)
UNITS mm	SCALE AS NOTED
BBOX 52.70 x 90.80 x 3.55 3rd-angle	SHEET 1/1